



Module-4: First Law of Thermodynamics

Thermodynamics (NMEC103) : (Winter 2024-25)

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For an ideal gas with constant specific heats

$$T_2 = T_1 + (V_1^2 - V_2^2)/2c_p$$

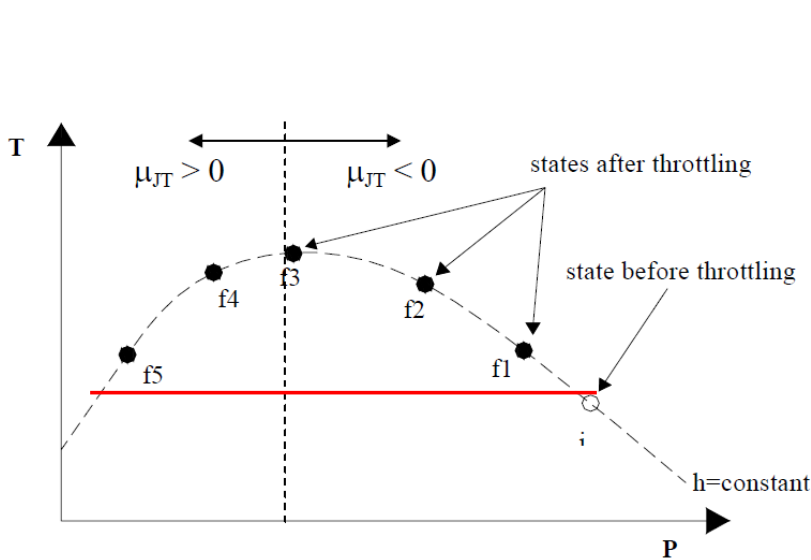
This equation tells that in the case of negligible change in specific kinetic energy that the throttling of an ideal gas the temperature of the gas before and after the throttling process will be same.

The actual throttling device outlet temperature for a pure substance depends upon the Joule-Thomson coefficient (an intensive property) defined as

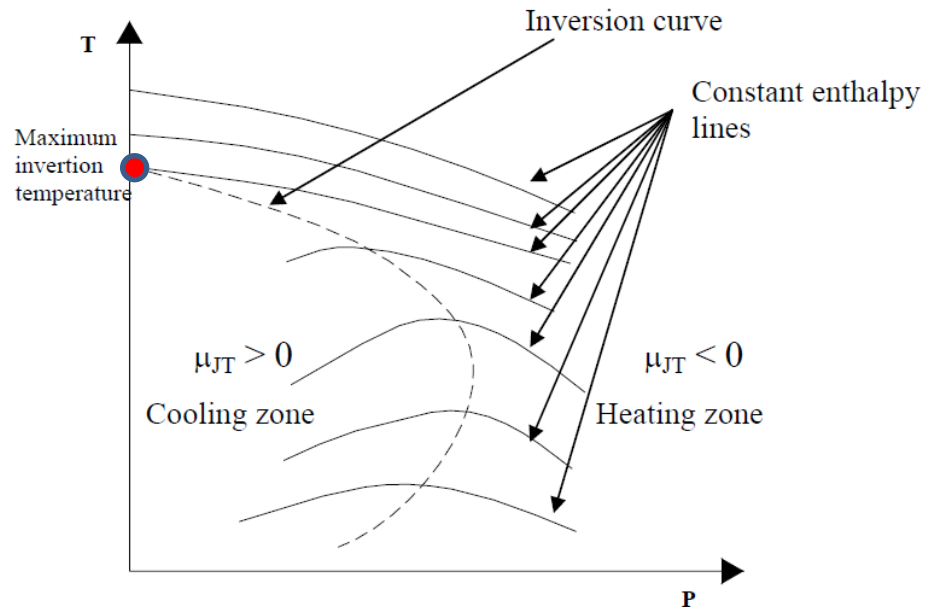
$$\mu_{JT} = \left(\frac{\partial T}{\partial p} \right)_h \approx (\Delta T / \Delta p)_{throttling}$$

Effect of Joule Thomson Expansion on Temperature

Joule Thomson Coefficient:
$$\mu_{JT} = \left(\frac{\partial T}{\partial P} \right)_h = - \left\{ v - T \left(\frac{\partial v}{\partial T} \right) \right\} / C_p$$



Isenthalpic expansion of a gas on T-P coordinates



Isenthalpic lines on T-P coordinates

At the Inversion Point: $\mu_{JT} = 0$

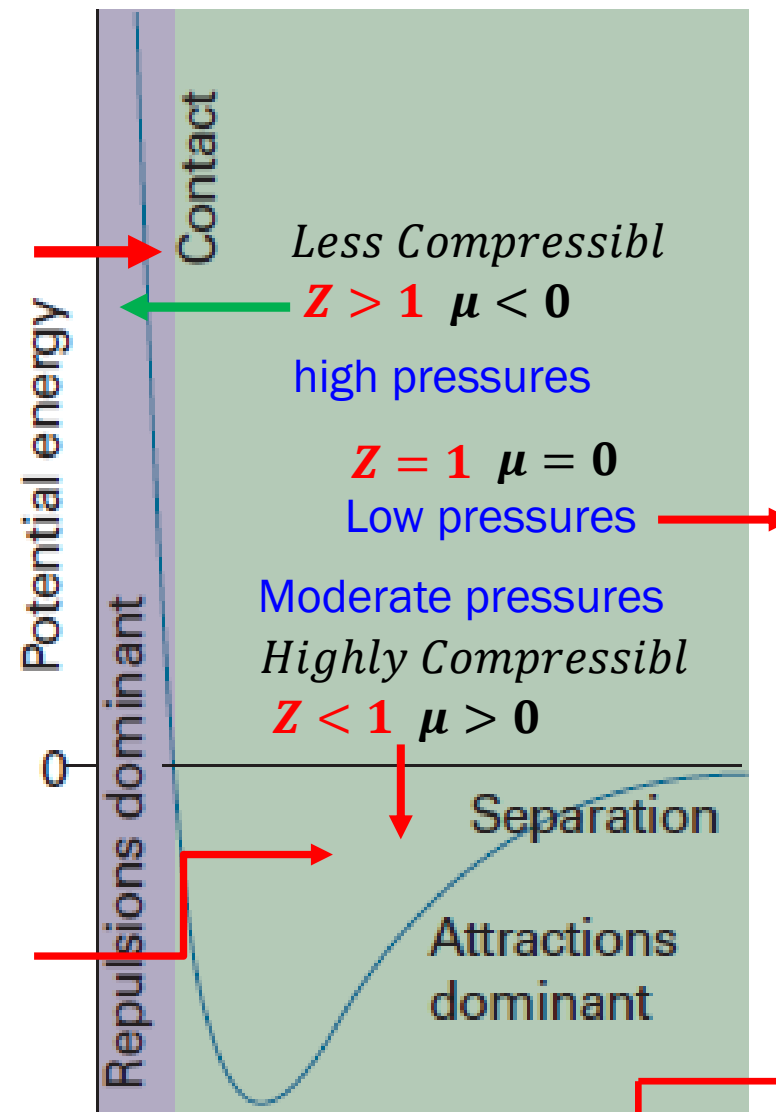
μ_{JT} is usually higher at low temperature

Joule Thomson Expansion of Real and Ideal Gases

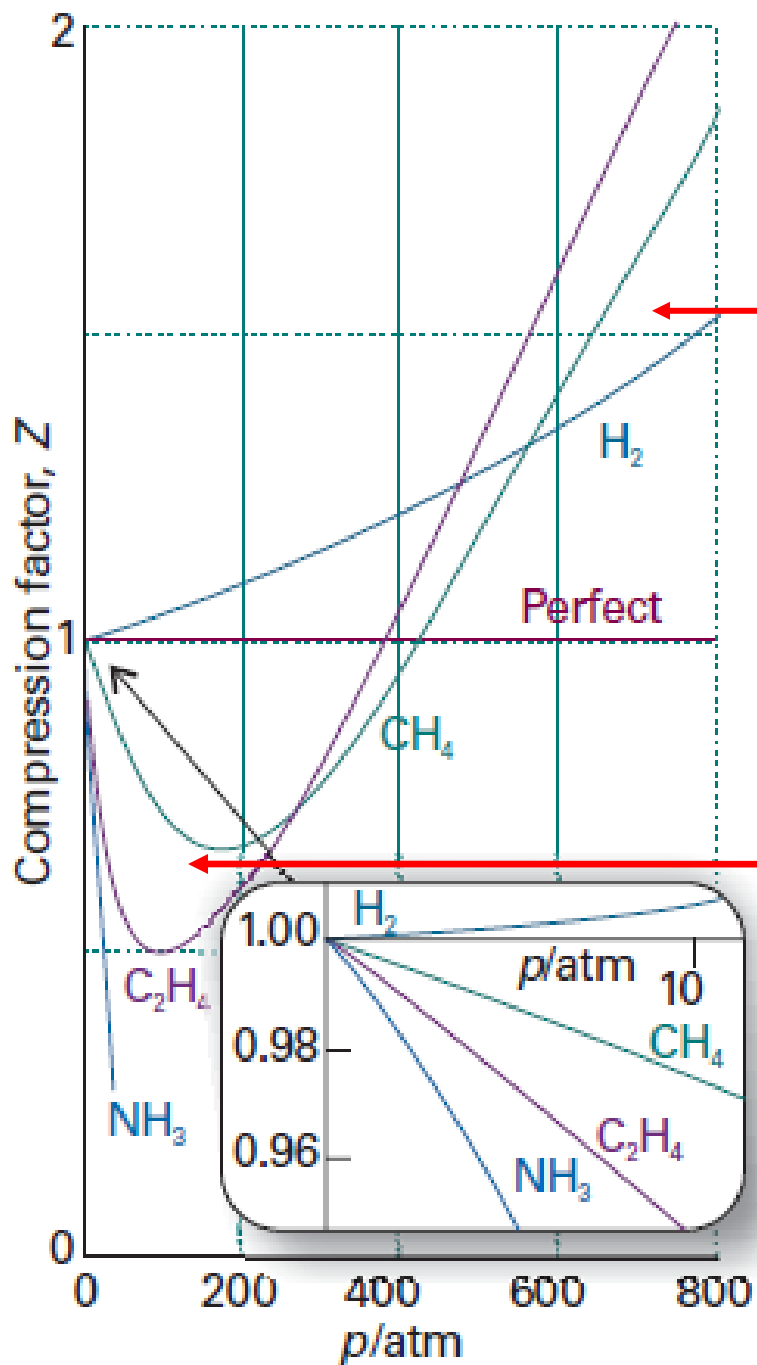
The kinetic model of gases and the equipartition theorem imply that the mean kinetic energy of molecules in a gas is proportional to the temperature. It follows that reducing the average speed of the molecules is equivalent to cooling the gas. If the speed of the molecules can be reduced to the point that neighbours can capture each other by their intermolecular attractions, then the cooled gas will condense to a liquid. To slow the gas molecules, we make use of an effect similar to that seen when a ball is thrown into the air: as it rises it slows in response to the gravitational attraction of the Earth and its kinetic energy is converted into potential energy. We saw that molecules in a real gas attract each other (the attraction is not gravitational, but the effect is the same). It follows that, if we can cause the molecules to move apart from each other, like a ball rising from a planet, then they should slow. It is very easy to move molecules apart from each other: we simply allow the gas to expand, which increases the average separation of the molecules. To cool a gas, therefore, we allow it to expand without allowing any energy to enter from outside as heat. As the gas expands, the molecules move apart to fill the available volume, struggling as they do so against the attraction of their neighbours. Because some kinetic energy must be converted into potential energy to reach greater separations, the molecules travel more slowly as their separation increases. This sequence of molecular events explains the Joule–Thomson effect: the cooling of a real gas by adiabatic expansion. The cooling effect, which corresponds to $\mu > 0$, is observed under conditions when attractive interactions are dominant ($Z < 1$), because the molecules have to climb apart against the attractive force in order for them to travel more slowly. For molecules under conditions when repulsions are dominant ($Z > 1$), the Joule–Thomson effect results in the gas becoming warmer, or $\mu < 0$.

Repulsive forces are significant only when molecules are almost in contact: they are short-range interactions, even on a scale measured in molecular diameters. Because they are short-range interactions, repulsions can be expected to be important only when the average separation of the molecules is small. This is the case at high pressure, when many molecules occupy a small volume. At high pressures, when the average separation of the molecules is small, the repulsive forces dominate and the gas can be expected to be less compressible because now the forces help to drive the molecules apart. ($Z > 1$)

On the other hand, attractive intermolecular forces have a relatively long range and are effective over several molecular diameters. They are important when the molecules are fairly close together but not necessarily touching (at the intermediate separations. At moderate pressures, when the average separation of the molecules is only a few molecular diameters, the attractive forces dominate the repulsive forces. In this case, the gas can be expected to be more compressible than a perfect gas because the forces help to draw the molecules together ($Z < 1$)



At low pressures, when the sample occupies a large volume, the molecules are so far apart for most of the time that the intermolecular forces play no significant role, and the gas behaves virtually perfectly. ($Z = 1$)

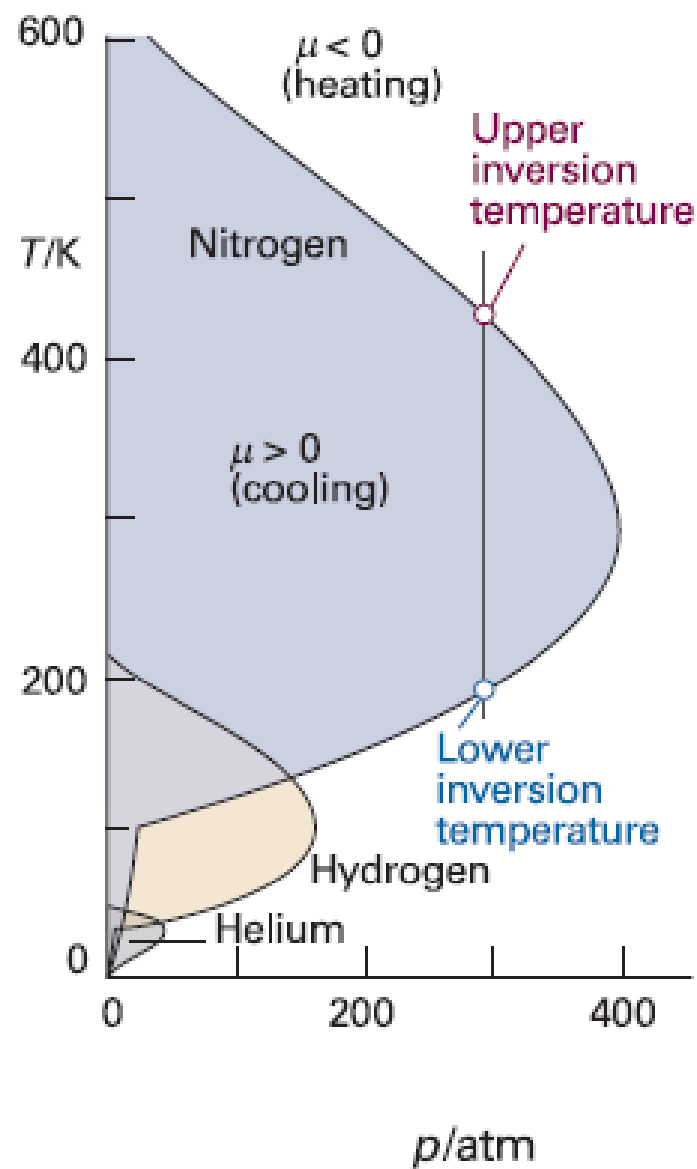
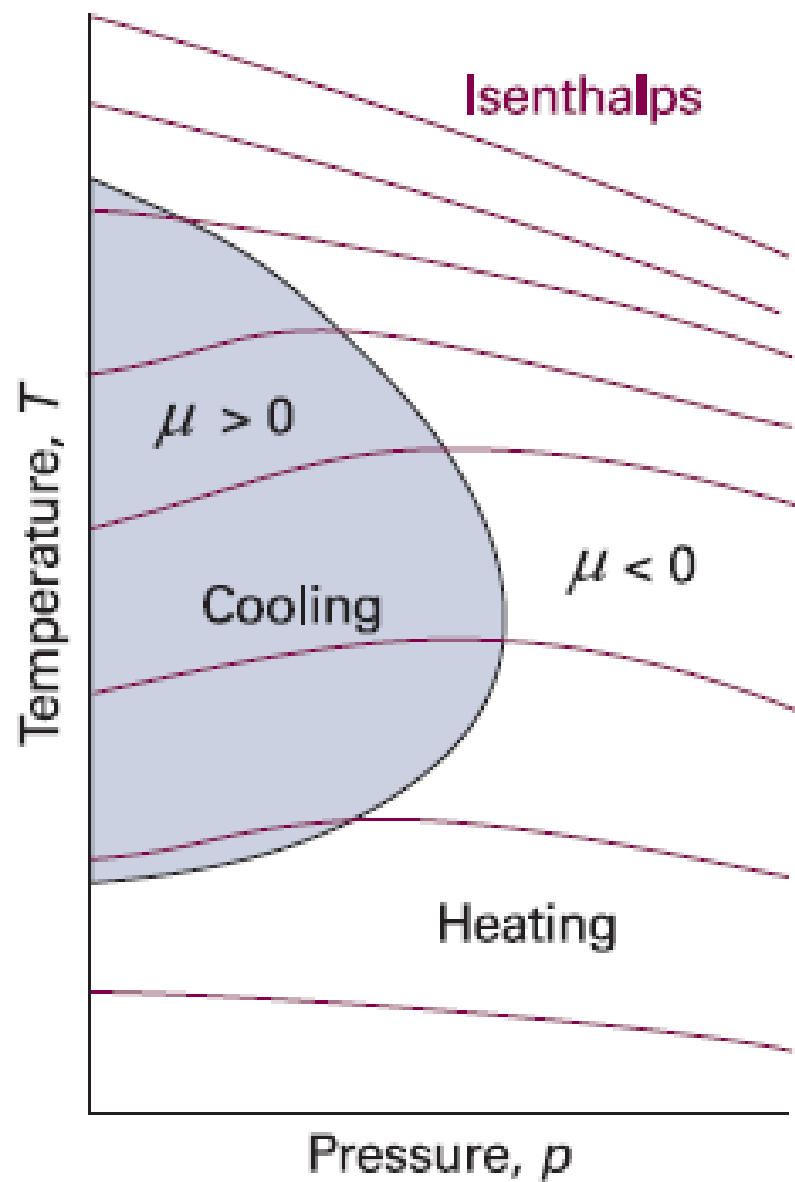


Less compressible

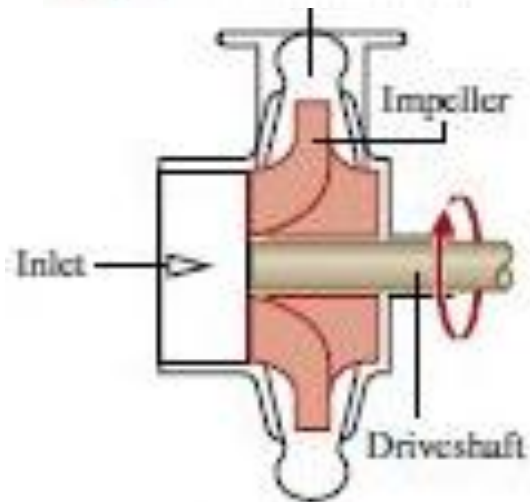
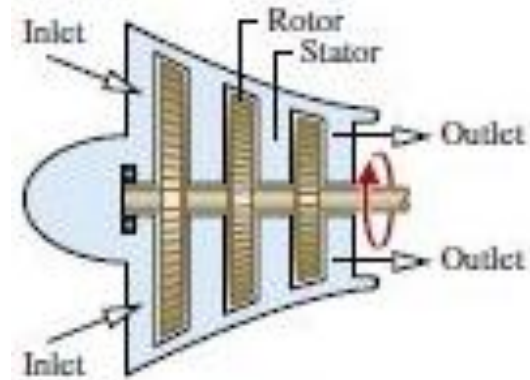
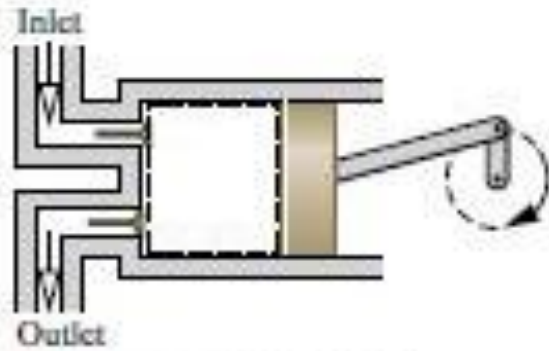
$$Pv = ZRT$$

$$\left(\frac{\partial v}{\partial P}\right)_T = -\frac{ZRT}{P^2}$$

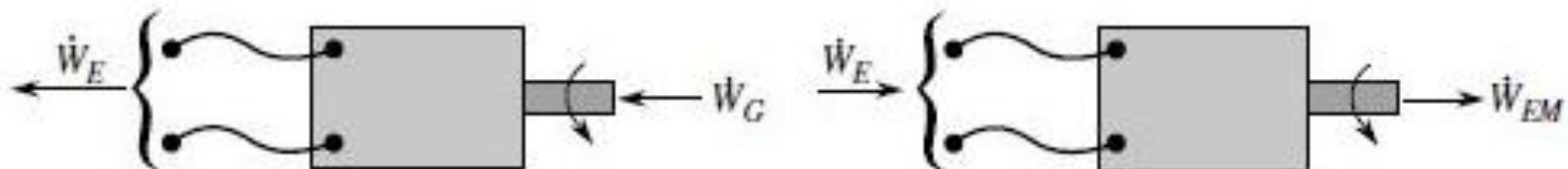
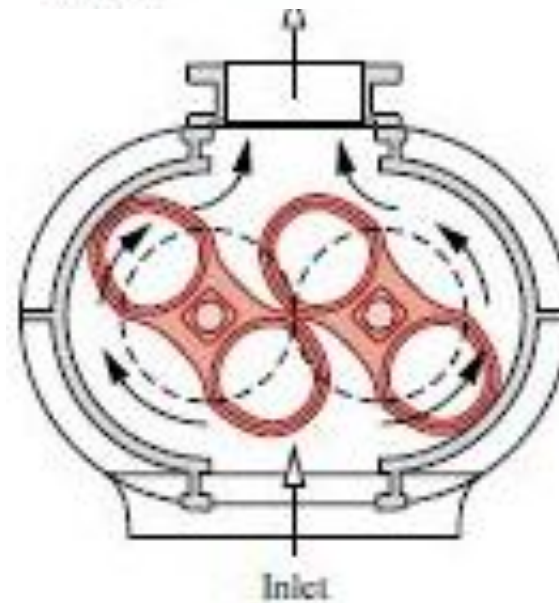
Highly compressible

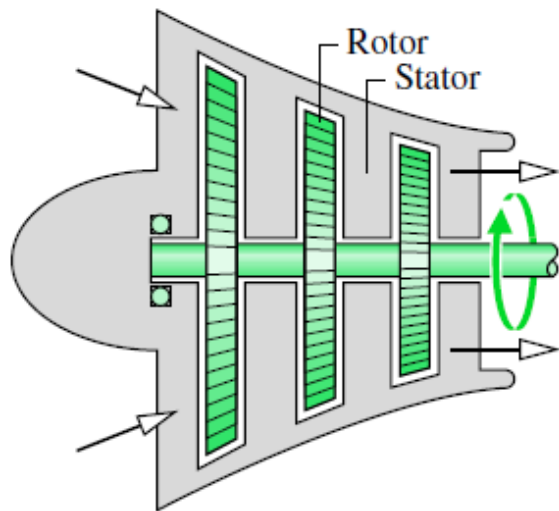


Shaft Work Machines: Turbines, Pumps/Compressors/motors/generators

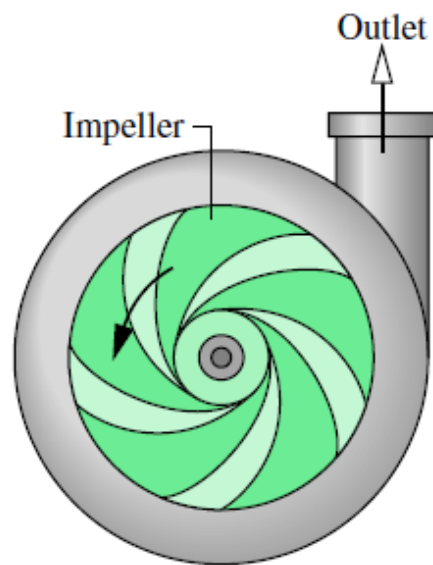


(c) Centrifugal

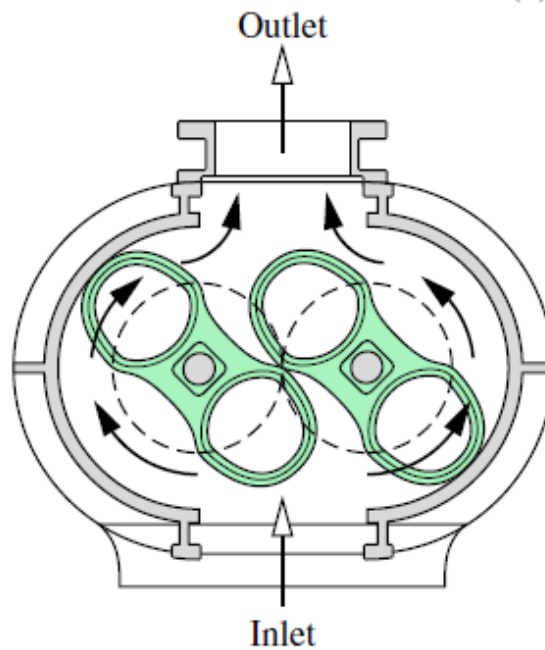
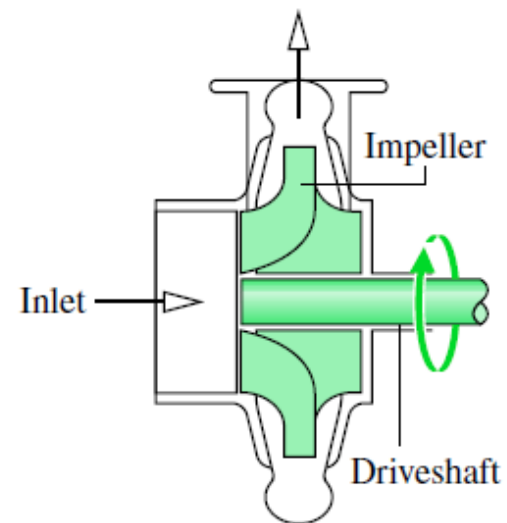




(a)



(b)



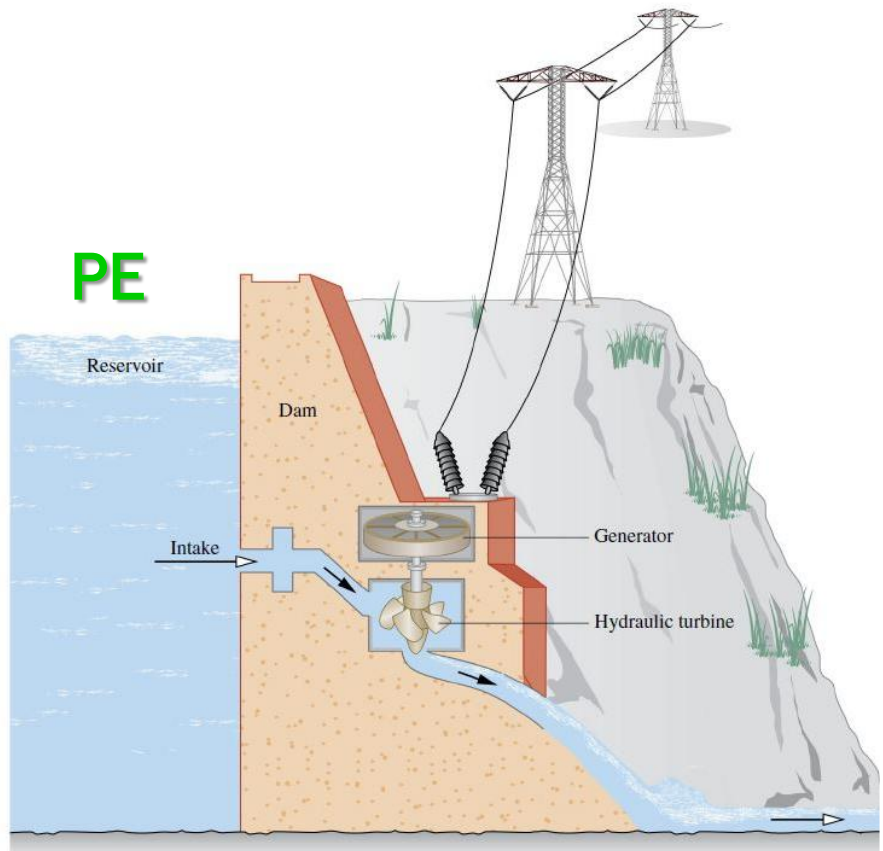
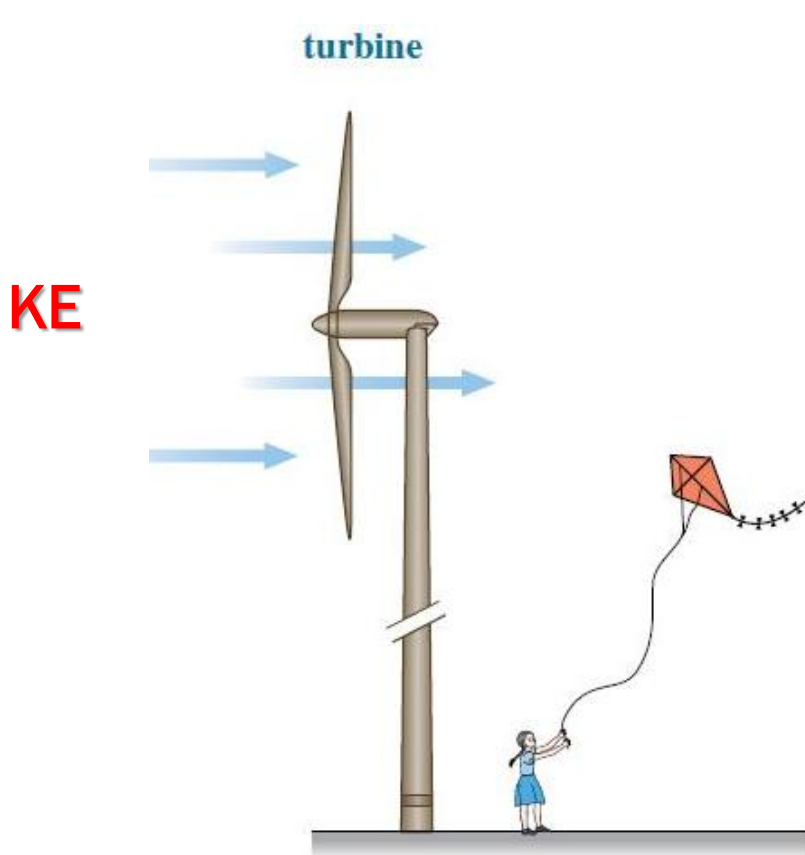
(c)

Shaft work machines are devices whose primary function is to promote a work input or output through a rotating or reciprocating shafts. Common shaft work machines are hydraulic pumps, pneumatic compressors, fans, gas or hydraulic turbines , electric motors and generators, external and internal combustion engines. Most shaft work machines are steady state. Steady flow , single inlet and single outlet devices (electric motors and generators are exceptions since they have no flow stream)

$$\dot{W}_{shaft} = \dot{Q} - \dot{m}(h_2 - h_1 + V_2^2/2 - V_1^2/2 + (Z_2 - Z_1)g)$$

This equation shows that the effect of heat loss from a work producing machine is to reduce the device's power output. Therefore, most work producing machines (engines and turbines) are insulated to improve the efficiency. Similarly heat loss from a work-absorbing device (such as compressor) will require more work be supplied to produce the same change in the flow streams.

Most shaft work machines have negligible change in specific kinetic and potential energies of their flow streams. Obvious exceptions are hydroelectric water turbines in which the specific potential energy change of the water is the energy source for the turbine, and the windmills in which the specific kinetic energy change of the air is the energy source for the wind mill.



For an insulated shaft work machine with negligible change in KE and PE of the flow streams, operating in a steady state, steady flow, single-inlet and single outlet condition is obtained from

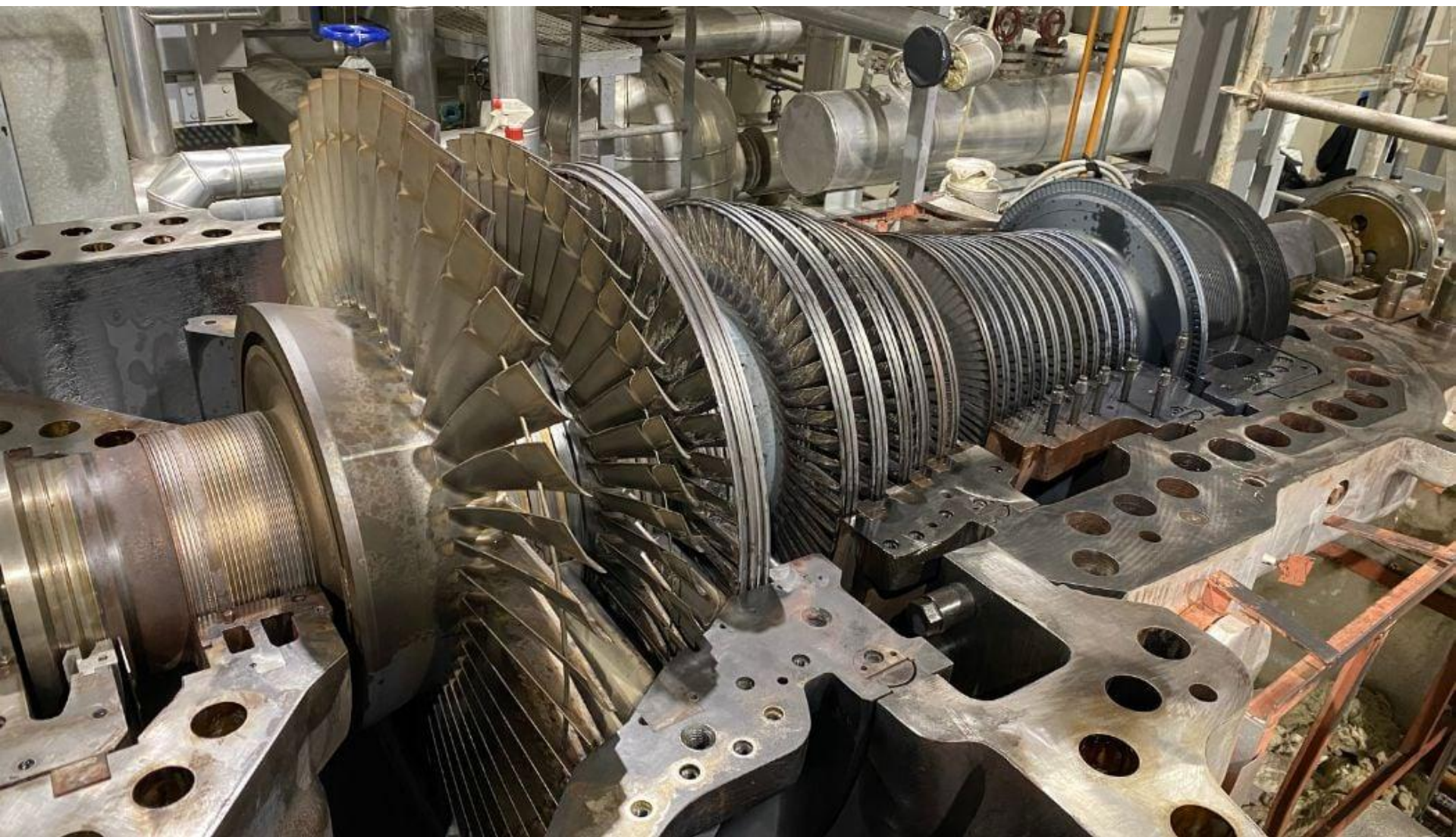
$$\dot{W}_{shaft} = \dot{m}(h_1 - h_2)$$

$$\dot{W}_{shaft, incomp. fluid} = \dot{m}(c(T_1 - T_2) + v(p_1 - p_2))$$

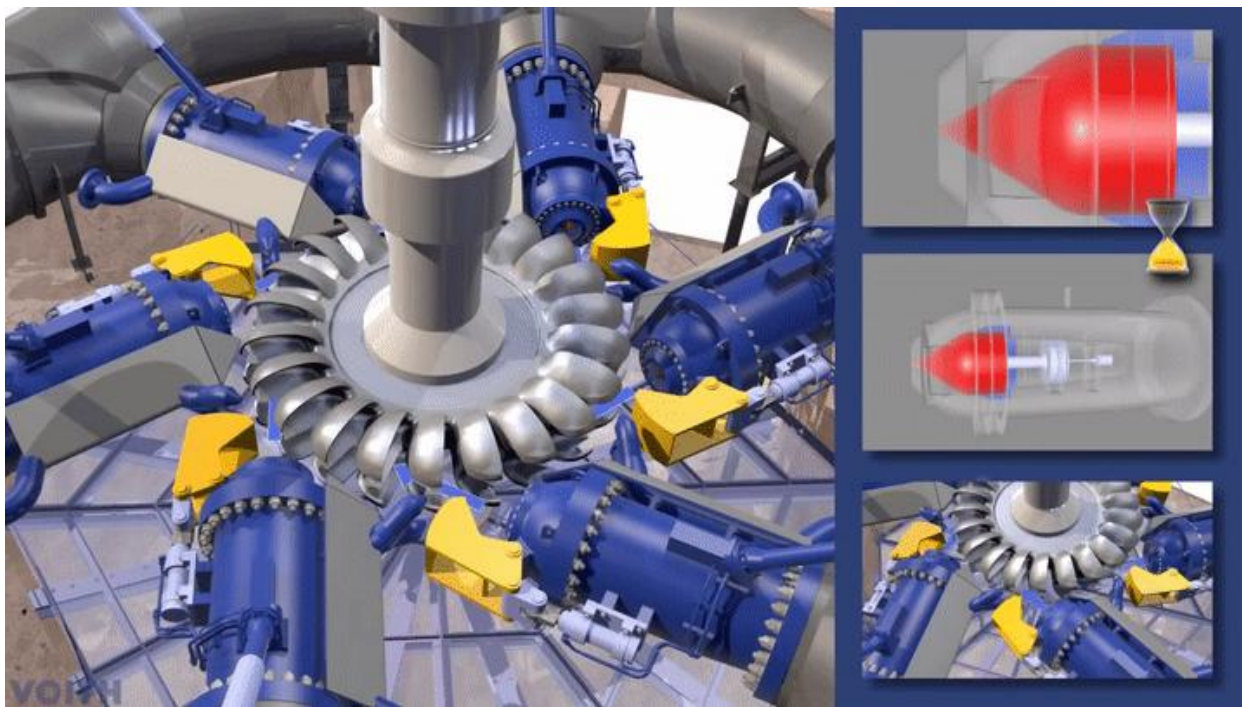
Normally there is very little change in temperature across such devices as hydraulic pumps, motors and turbines

$$\dot{W}_{shaft, isothermal incomp. fluid} = \dot{m}v(p_1 - p_2)$$

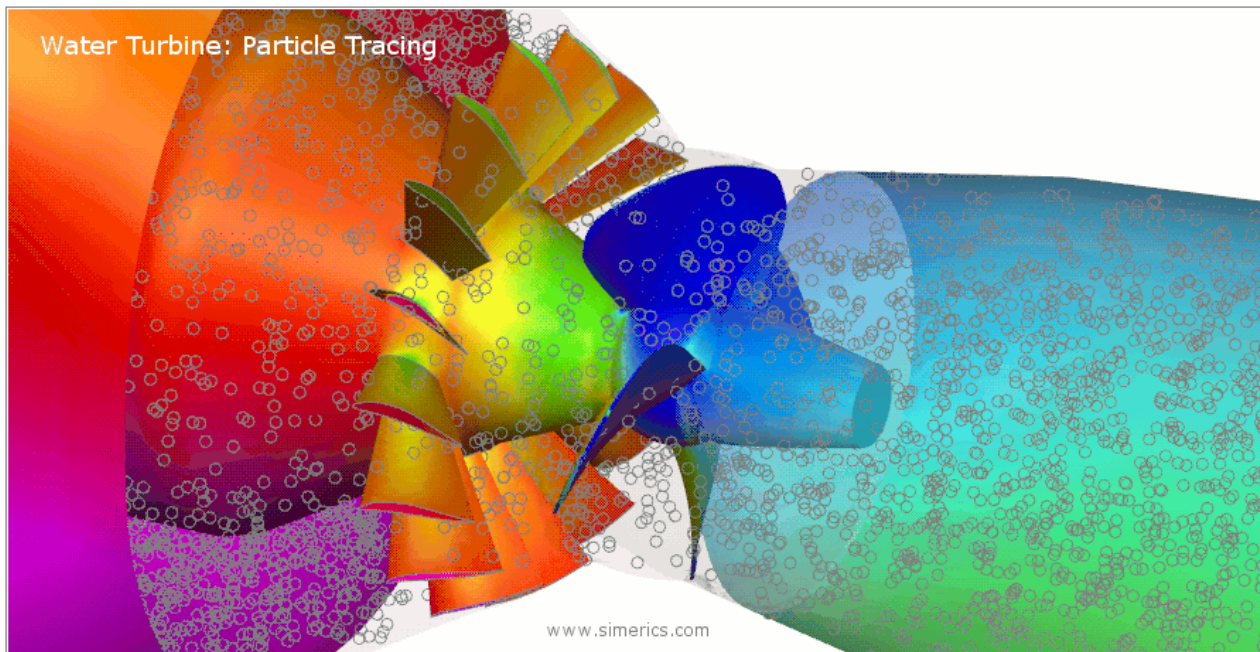
$$\dot{W}_{shaft, calorifically perfect gas} = \dot{m}c_p(T_1 - T_2)$$

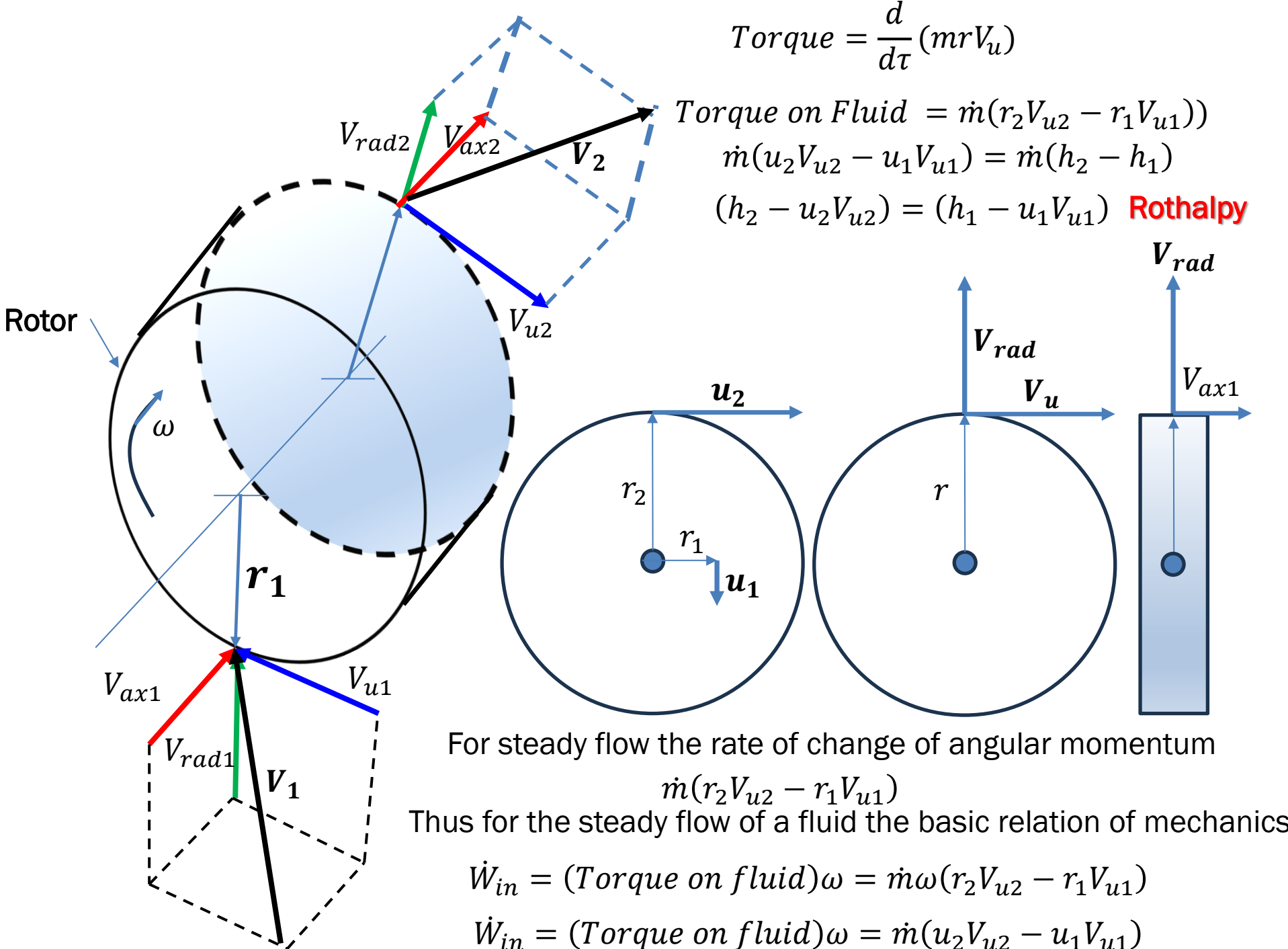


Steam Turbine



Water Turbine: Particle Tracing





Basic thermodynamics of turbines (contd.)

- **Stagnation state:**
- “It is the **hypothetical, terminal state** of a **fictitious, isentropic, work-free** and **steady flow process** during which the macroscopic **kinetic and potential energies** of the fluid are **reduced to zero**”
- The macroscopic kinetic and potential energies are defined with reference to an arbitrary reference or datum state
- From 1st law of thermodynamics. for the stagnation state:

$$q_{1-o} - w_{1-o} = h_o - \left(h_1 + \frac{V_1^2}{2} + gz_1 \right) = 0$$

$$\Rightarrow h_o = \left(h_1 + \frac{V_1^2}{2} + gz_1 \right)$$

$$(h_{o1} - u_1 V_{u1}) = (h_{o2} - u_2 V_{u2})$$

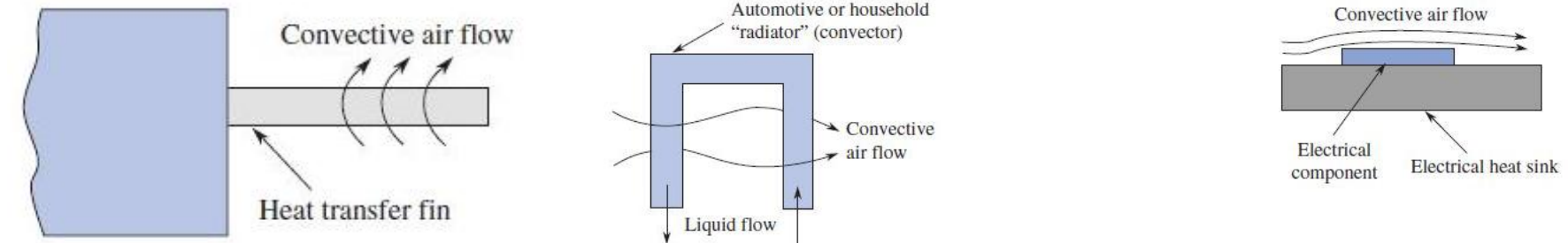
If the change in kinetic and potential energy are important across a turbine then

defining $(h_o - uV_u) = \text{Rothalpy, I}$

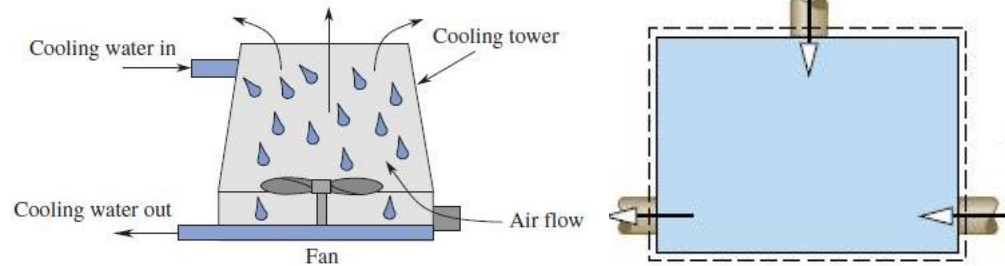
Heat Exchangers

A heat exchanger is a device which facilitates the heat transfer from a solid to fluid or from a fluid to another fluid with and without any intervening solid surface

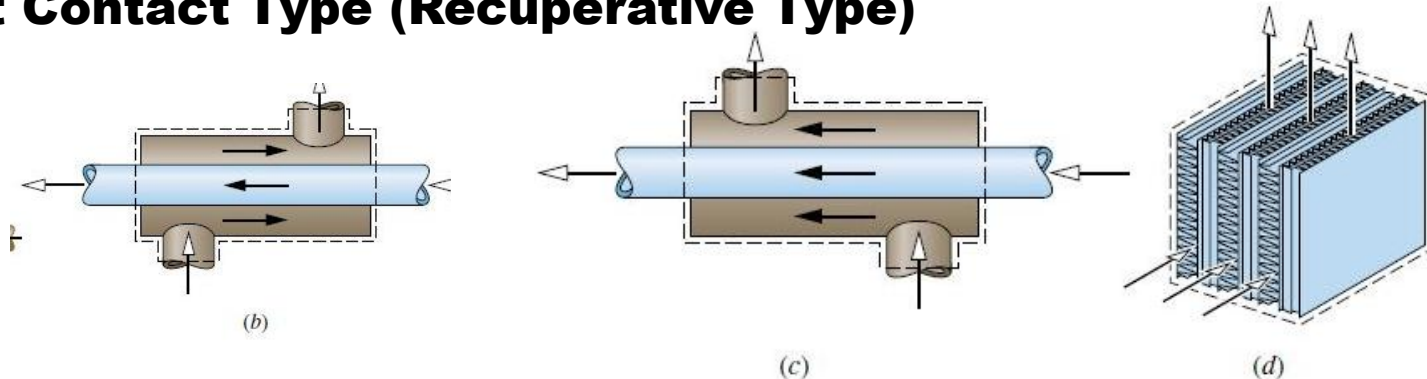
Solid to Fluid



Direct Contact type

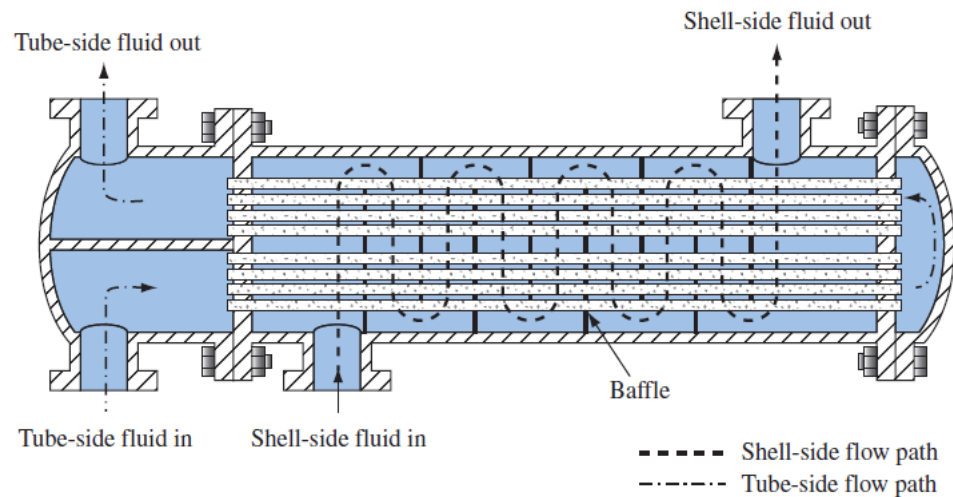


Indirect Contact Type (Recuperative Type)

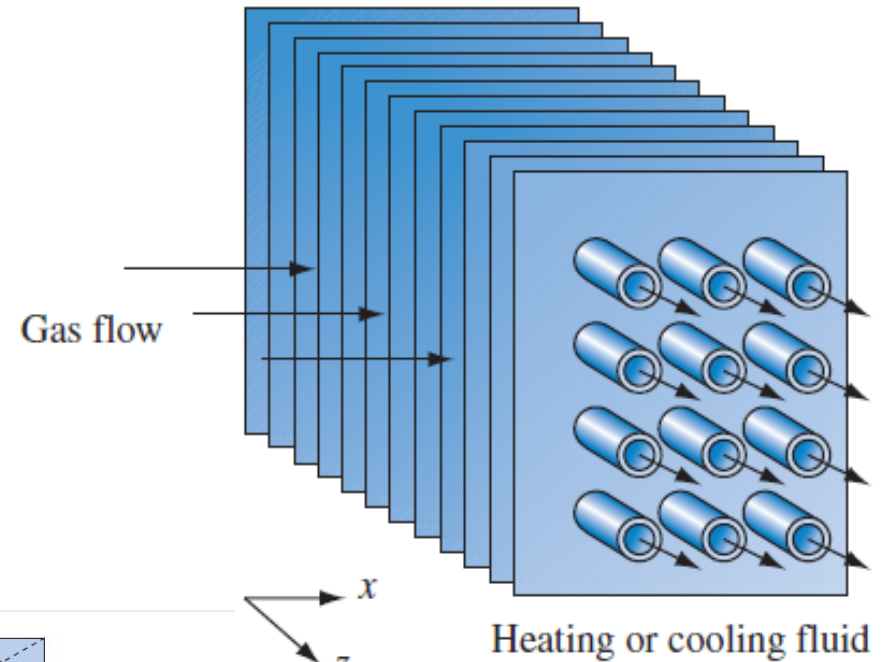
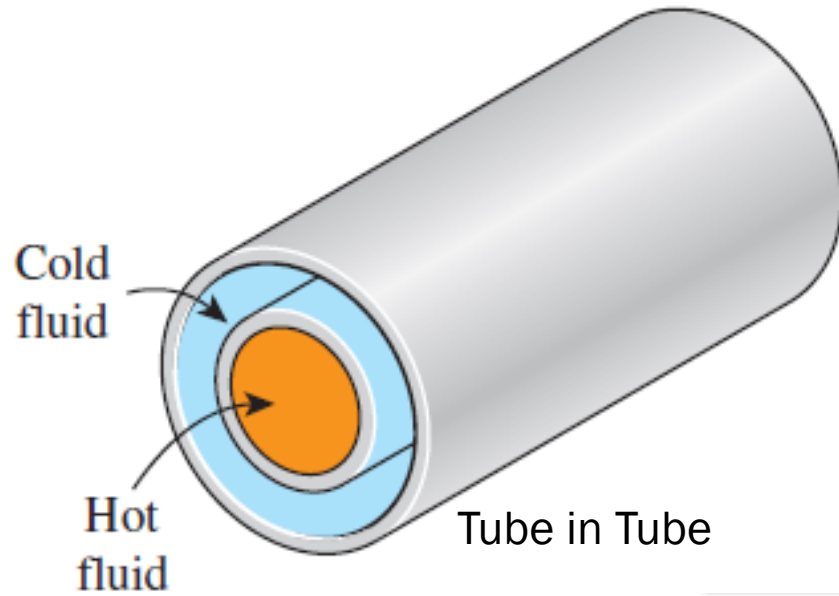
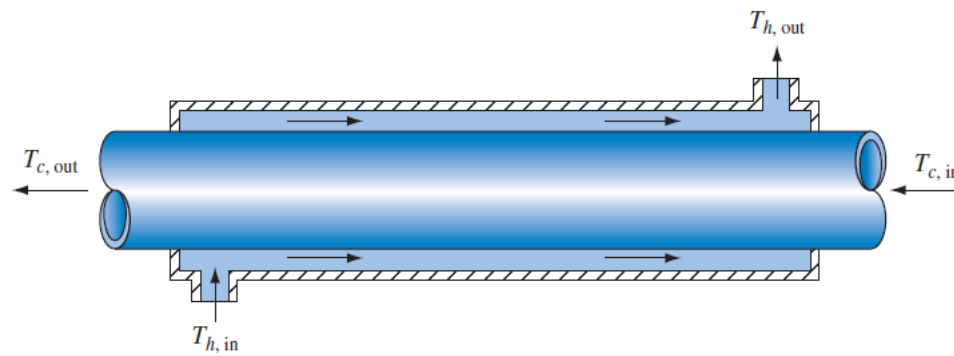




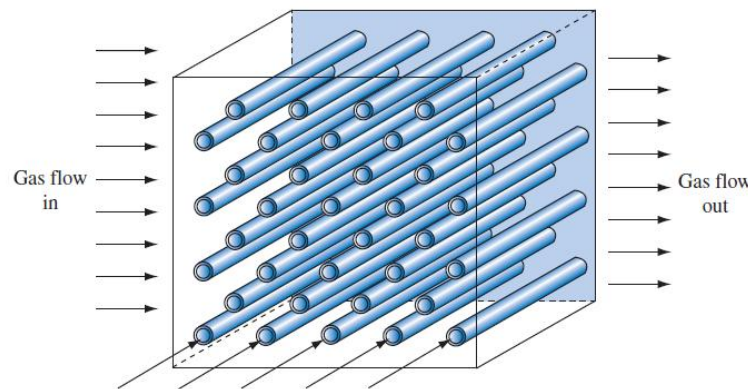
Recuperative Heat Exchanger (Both the hot and cold fluid flow simultaneously under goes energy interaction)



Shell and Tube Type Heat Exchanger



Fin & Tube HX
both side
unmixed



Cross flow one
side mixed



Spray Pond



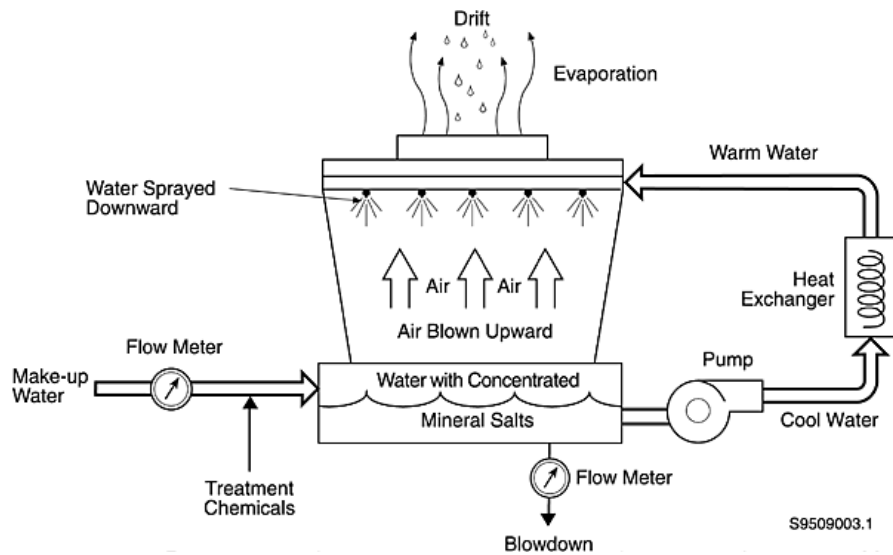
Cooling Pond



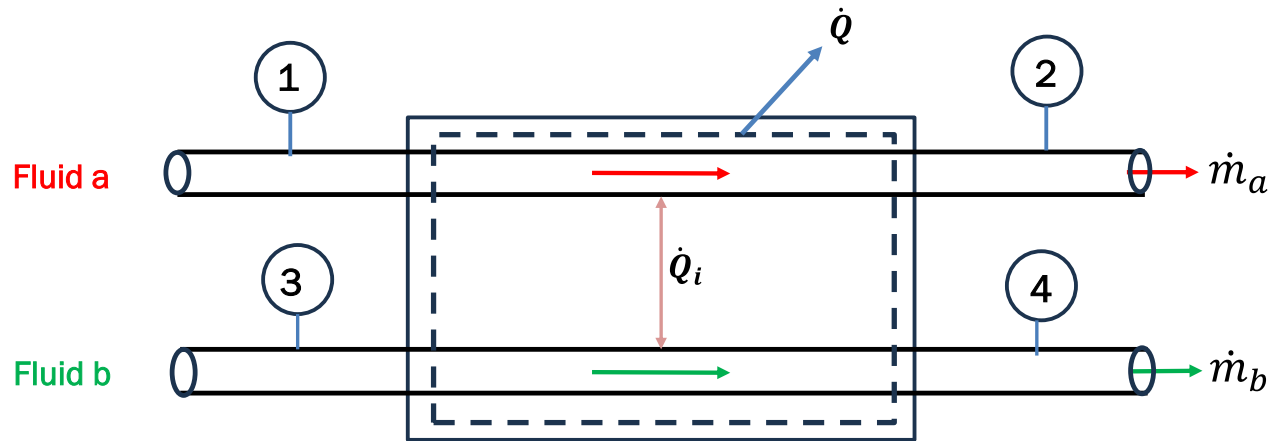
Cooling Towers



Spray Canal



Hyperbolic Cooling tower installed
in Westfalen power plant in
Germany



Applying steady flow energy equation neglecting the change in kinetic and potential energy assuming no scope for work interaction

$$\dot{Q} + \dot{m}_1 h_1 + \dot{m}_3 h_3 - \dot{m}_2 h_2 + \dot{m}_4 h_4 = 0$$

From mass conservation:

$$\dot{m}_1 = \dot{m}_2 = \dot{m}_a$$

$$\dot{m}_3 = \dot{m}_4 = \dot{m}_b$$

$$\dot{Q} + \dot{m}_a (h_1 - h_2) + \dot{m}_b (h_3 - h_4) = 0$$

If the heat exchanger is insulated from outside:

$$\dot{m}_a (h_1 - h_2) = \dot{m}_b (h_4 - h_3)$$

If both the fluid are assumed to be **almost constant density fluids**

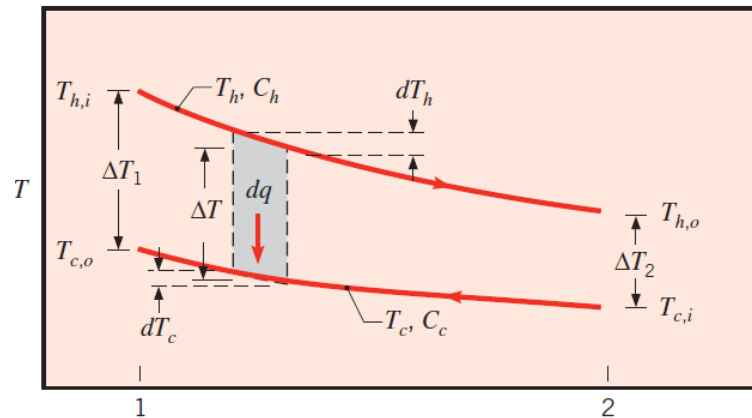
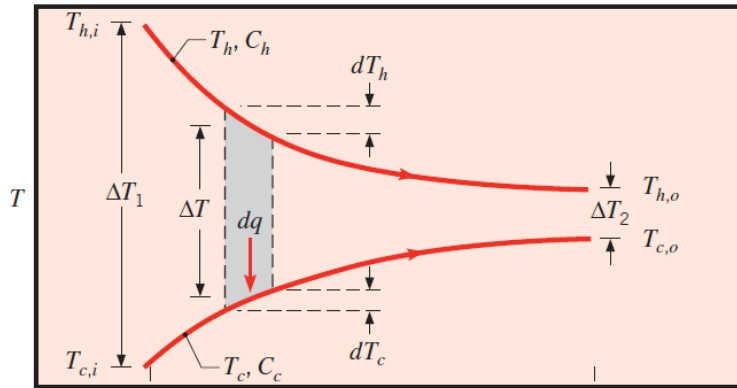
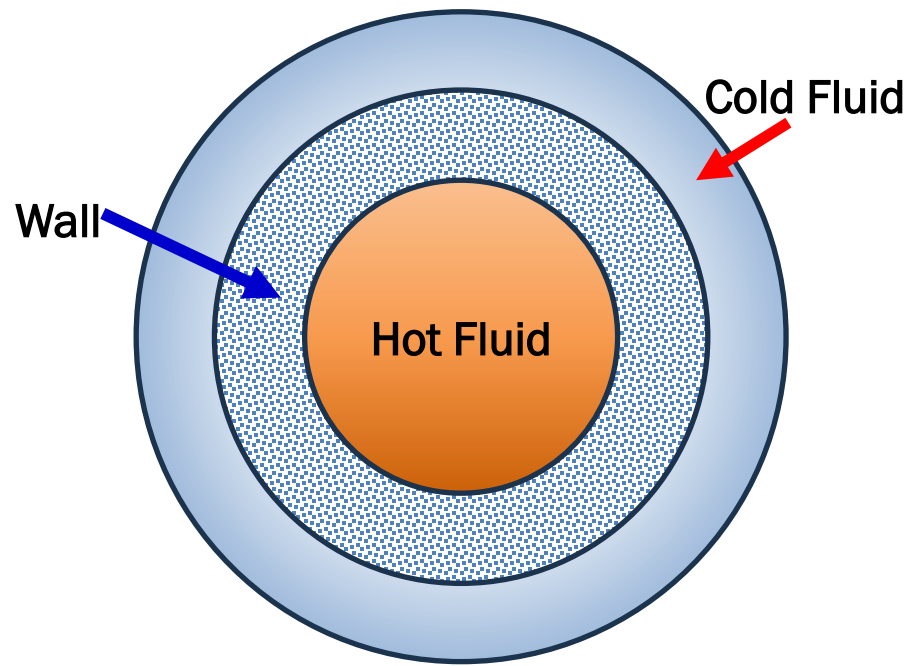
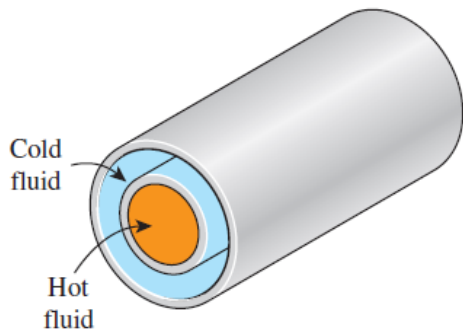
$$\dot{m}_a [C_a (T_1 - T_2) + v_a (p_1 - p_2)] = \dot{m}_b [C_b (T_4 - T_3) + v_b (p_4 - p_3)]$$

For **liquids** v_a and v_b are very small moreover the **pressure drops** are also negligible

$$\dot{m}_a [C_a (T_1 - T_2)] = \dot{m}_b [C_b (T_4 - T_3)]$$

If both the fluids are **ideal gases** (Considering average Specific Heat)

$$\dot{m}_a [C_{p,a} (T_1 - T_2)] = \dot{m}_b [C_{p,b} (T_4 - T_3)]$$



$$(\Delta T)_{\log mean} = \frac{\Delta T_2 - \Delta T_1}{\ln\left(\frac{\Delta T_2}{\Delta T_1}\right)}$$

$$q \rightarrow T_{h,f} \xrightarrow{R_{con}} T_{w,in} \xrightarrow{R_{cond}} T_{w,out} \xrightarrow{R_{con}} T_{c,f} \rightarrow q$$

$$\frac{1}{h_{in}A_{in}} \quad \frac{\ln\left(\frac{A_{out}}{A_{in}}\right)}{2\pi Lk_w} \quad \frac{1}{h_{out}A_{out}}$$

$$\frac{1}{UA} = \frac{1}{R_{con}} + \frac{1}{R_{cond}} + \frac{1}{R_{con}}$$

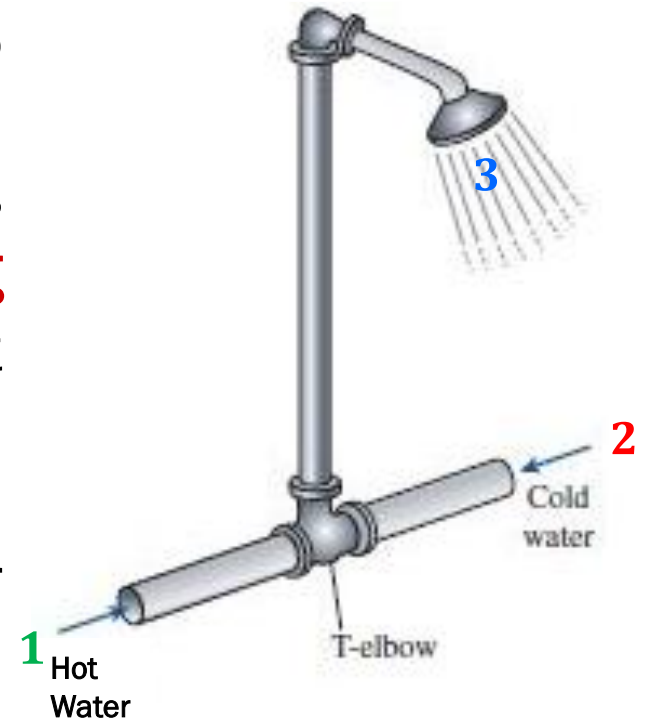
$$\dot{Q} = UA(\Delta T)_{\log mean}$$

$$\dot{m}_h(h_{in} - h_{out}) = \dot{m}_c(h_{out} - h_{in}) = UA(\Delta T)_{\log mean}$$

Overall Heat Transfer Coefficient/ Overall Conductance: U (W/m²-K)

Mixing Chambers

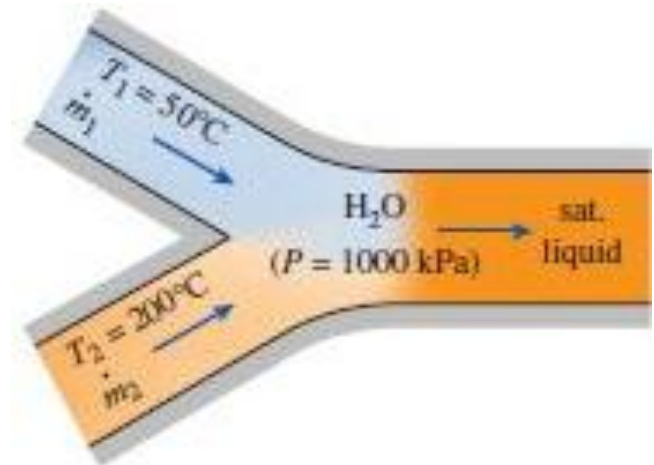
In engineering applications, mixing two streams of fluids is not a rare occurrence. The section where the mixing process takes place is commonly referred to as a **mixing chamber**. The mixing chamber does not have to be a distinct “chamber.” An ordinary T-elbow or a Y-elbow in a shower, for example, serves as the mixing chamber for the cold- and hot-water streams



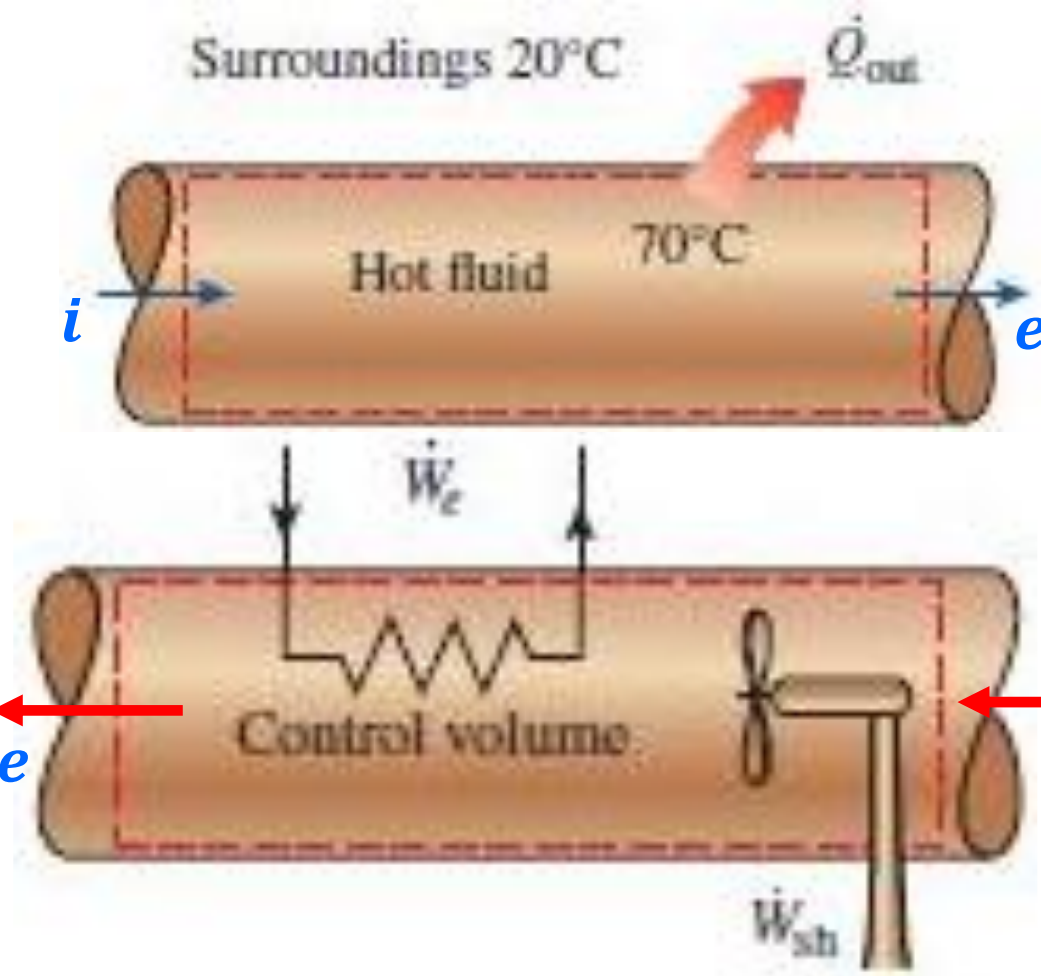
$$\dot{W} = 0; \dot{Q} \approx 0; \Delta KE \approx 0; \Delta PE \approx 0$$

$$\dot{m}_1 + \dot{m}_2 = \dot{m}_3$$

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = (\dot{m}_1 + \dot{m}_1) h_3$$



Flow through Pipes and Ducts



$$-\dot{Q}_{out} = \dot{m}(h_e - h_i)$$

If the **pressure drop** is negligible the approximate **thermodynamic process** is **isobaric**

$$-\dot{W}_{sh} - \dot{W}_e = \dot{m} \left(h_e - h_i + \frac{V_e^2}{2} - \frac{V_i^2}{2} \right)$$